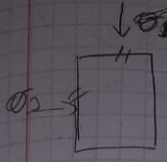
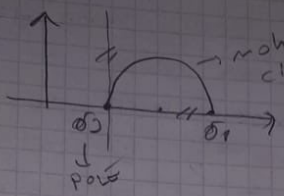
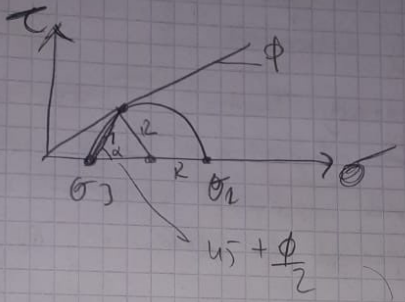

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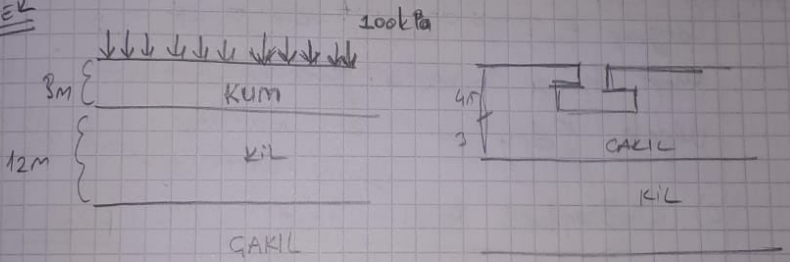


Coulomb's equation

$\tau = c + \sigma \tan\varphi$



ÖRNEK



COMPACTION (İYİLEŞTİRME - soil improvement method)

Compaction is a type of compression. Compaction should not be confused with another type of compression (consolidation).

Compaction occurs due to the decrease in voids occupied by air. Consolidation occurs due to the decrease in voids occupied by water. In addition, compaction occurs immediately (just after compaction load is applied on soils).

consolidation occurs many years later load is applied on soils. Besides, according to the concept of compaction, soil is not completely saturated but according to the concept of consolidation soil is completely saturated

Laboratory Tests, There are two laboratory tests for compaction. These are standard proctor test and modified proctor test. Both tests are performed to determine $\gamma_{dry\,max}$ and w_{opt}

$\gamma_{dry\,max}$ = max dry unit weight of soil

w_{opt} is a special water content at which γ_{dry} will be maximum

is commonly used all over the world

Standard Proctor Test

A mold (950cm³) at a certain water content soil is placed into the mold in three layers

A standard weight (2.5kg) is dropped onto the each layer 25 times From a standart height of 30cm

Modified Proctor Test

A mold (950 cm³) at a certain water content soil is placed in the mold in five layers

A standard weight (4.5kg) is dropped onto the each layer 25 times From a standart height of 45cm

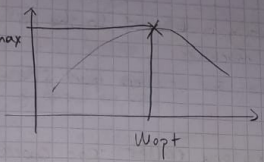
This test is
Contents

$\gamma_{dry\,max}$

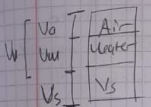
$\gamma_{dry} = \gamma$

Va Air
Vw Water
Vs

This test is repeated at different water contents



$$\gamma_{dry} = \frac{\gamma_{wet}}{1+w} = \frac{\gamma_{moist}}{1+w} \quad (1)$$



$$\frac{V_T}{V_s} = \frac{V_w}{V_s} + \frac{V_s}{V_s} \Rightarrow V_T = V_s[1+e]$$

$$V_T = V_s(1+e) \quad (2)$$

W_s is constant

V_s is constant for soils (3)

$$\gamma_s = \frac{W_s}{V_s} \text{ is constant}$$

Compaction is performed in fields. If compaction is performed on an existing soil layer in the field, compaction will be succesfull at a depthe of up to 2m - 10m depending on the type of compaction energy. If compaction is performed on an excavated soil to construct a compacted fill, compaction will be succesfull for the height of the fill.

Compaction energy is delivered to the soil by using some special machines

- Sheepsfoot rollers for fine-grained soil (5–10 kez)
- Vibratory rollers for coarse-grained soil (10–15 kez)

For the process, it is important to determine the thickness of layers (lift) which will be placed the number of passing of machine the water content which fill is compacted

NOTE: Dynamic consolidation (Dynamic compaction)

100 kN – 400 kN (10t – 40t)

↓ = 10m – 40m bırakılıyor bu şekilde

(+) Bu yöntem
dezavantajları
derin gibi titreşimler oluşturur
Çevresine ve zemin sıvılaşmasına sebep olur

(+) Elebia'taki en iyi zemin kompakte edilmiş zemindir

(+) Kazılmış zeminde işçi hatasıyla kompakte olur

Control of co

these methods are
on determining
fill

$$\frac{\gamma_n}{1+w}$$

bepree of

$$= \frac{\gamma_{dry}}{\gamma_{ds}}$$

* kompaksiyo
detail malzem
kayalık
fullanışlı

Example

A compacted
a silty cl
be excavat
transported
excavation
γ_{wet} = 1
The resul
performed
excavation
RC = %

Control of compaction: Sand Cone Method
Rubber Balloon method
Nuclear Method
*En çok kullanılan metot

these methods are based on determining the volume of a hole dug in the fill

$\frac{\gamma_n}{1+w} = \gamma_{dry}$ bulunur bu metodlarla

Degree of compaction = RC = Relative Compaction

$= \frac{\gamma_{dry\ site}}{\gamma_{dry\ max\ lab}}$ (RC must be min %95)

* Kompaksiyon yerinde sıkıştırma için çok zaman alır depil malzeme: aynı yerine kaz ayesela alttan 2m kayalık sert kat pitsin ama dolgu için çok tavsiye edilmiştir.

Example

A compacted fill be constructed by using a silty clay soil. This silty clay soil will be excavated from a site and will be transported to a construction site. At the excavation site (borrow site = borrow pit) $\gamma_{wet} = 12 kN/m^3$, $w = 0.1$ and $G_s = 2.7$

The results of a standard proctor test performed on a soil sample taken from excavation site were tabulated below

RC = %100

During transportation, %5 of loss of water content has been assumed to occur

γ_{wet} (kN/m³)	17.7	19.36	21.66	20.88
w %	8	10	14	16

a-) Plot γ_{dry} - w curve

b-) determine γ_{dry} and w_{opt}.

c-) calculate V_{fill} if 20.000 m³ of soil is excuated from the borrow pit

d) Calculate Δw to be added into fill

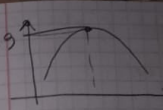
a-) $\gamma_{dry} = \frac{17.7}{1+0.08} = 16.4$ kN/m³

$\gamma_{dry} = \frac{19.36}{1+0.1} = 17.6$ kN/m³

$\gamma_{dry} = \frac{21.66}{1+14} = 19$ kN/m³

$\gamma_{dry} = \frac{20.88}{1.16} = 18$ kN/m³

$\gamma_{dry} = \frac{20.05}{1.18} = 17.5$ kN/m³



c-) $\gamma_{dry\,exc}$

10. 9

%100 Re

γ_9 =

20000

$V_{exc} = V_s$

$V_{fill} = V_s$

20000

V_{sfill}

d) $w_{exc} = \frac{V}{V}$

$W_{w_{exc}}$ = 327

32700 -

w_{fill} =

Δw =



b-) $\gamma_{dry\ max} = 19\ kN/m^3$

$w_{opt} = \%14$

c-) $\gamma_{dry\ exc} = \frac{12}{1+0.1} = 10.9\ kN/m^3$

$10.9 = \frac{W_{s\ exc}}{30000} \Rightarrow W_{s\ exc} = 32700\ kN$

%100 Relatif kompaksiyon $\gamma_{FILL} = 19$

$19 = \frac{327000\ kN}{V_{total\ fill}} = 17200 m^3$

$\frac{30000}{V_{sFill}} = \frac{1+e_{exc}}{1+0.42} \Rightarrow V_{Fill} = 17170 m^3$

$V_{exc} = V_s(1+e_{exc}) \rightarrow 1.48$

$V_{Fill} = V_s(1+e_{fill}) \rightarrow 0.42$

$\gamma_{dry_{exc}} = \frac{\gamma_s}{1+e_{exc}} = \frac{G_s \gamma_w}{1+e_{exc}}$

$10.9 = \frac{G_s \gamma_w}{1+e_{exc}}$

d) $w_{exc} = \frac{W_{w\ exc}}{W_s} = 0.1 = \frac{W_w}{327000\ kN} \Rightarrow e_{exc} = 1.48$

$W_{w_{exc}} = 32700\ kN$

$32700 - 327000\left(\frac{5}{100}\right) = 31065\ kN \Rightarrow e_{fill} = 0.42$

$w_{fill} = 0.14 = \frac{W_{w\ fill}}{32700} \Rightarrow W_{w\ fill} = 45780\ kN$

$\Delta W = 14715\ kN$ is added into the fill

$\Delta V_w = 1476 m^3$